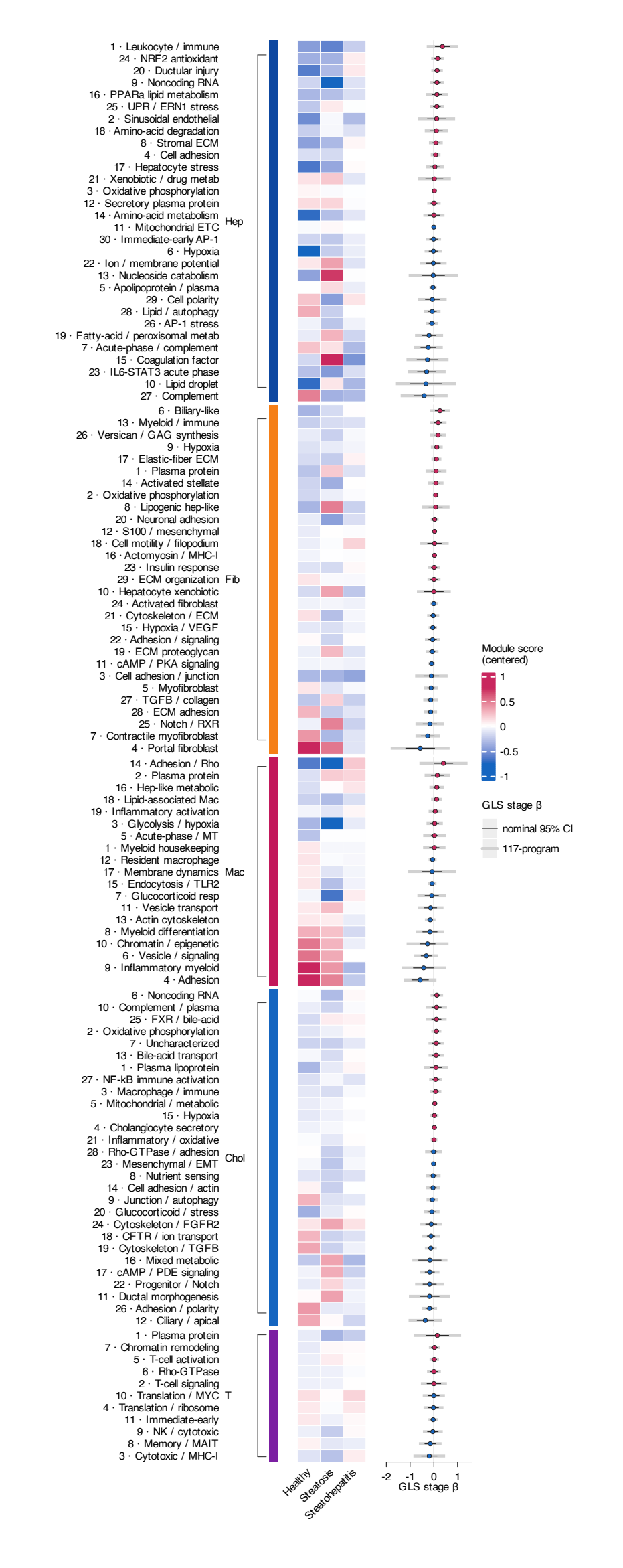
1 · Leukocyte / immune
24 · NRF2 antioxidant
20 · Ductular injury
9 · Noncoding RNA
16 · PPARa lipid metabolism
25 · UPR / ERN1 stress
2 · Sinusoidal endothelial
18 · Amino-acid degradation
8 · Stromal ECM
4 · Cell adhesion
17 · Hepatocyte stress
21 · Xenobiotic / drug metab
3 · Oxidative phosphorylation
12 · Secretory plasma protein
14 · Amino-acid metabolism
11 · Mitochondrial ETC
30 · Immediate-early AP-1
6 · Hypoxia
22 · Ion / membrane potential
13 · Nucleoside catabolism
5 · Apolipoprotein / plasma
29 · Cell polarity
28 · Lipid / autophagy
26 · AP-1 stress
19 · Fatty-acid / peroxisomal metab
7 · Acute-phase / complement
15 · Coagulation factor
23 · IL6-STAT3 acute phase
10 · Lipid droplet
27 · Complement
Hep
6 · Biliary-like
13 · Myeloid / immune
26 · Versican / GAG synthesis
9 · Hypoxia
17 · Elastic-fiber ECM
1 · Plasma protein
14 · Activated stellate
2 · Oxidative phosphorylation
8 · Lipogenic hep-like
20 · Neuronal adhesion
12 · S100 / mesenchymal
18 · Cell motility / filopodium
16 · Actomyosin / MHC-I
23 · Insulin response
29 · ECM organization
10 · Hepatocyte xenobiotic
24 · Activated fibroblast
21 · Cytoskeleton / ECM
15 · Hypoxia / VEGF
22 · Adhesion / signaling
19 · ECM proteoglycan
11 · cAMP / PKA signaling
3 · Cell adhesion / junction
5 · Myofibroblast
27 · TGFB / collagen
28 · ECM adhesion
25 · Notch / RXR
7 · Contractile myofibroblast
4 · Portal fibroblast
Fib
14 · Adhesion / Rho
2 · Plasma protein
16 · Hep-like metabolic
18 · Lipid-associated Mac
19 · Inflammatory activation
3 · Glycolysis / hypoxia
5 · Acute-phase / MT
1 · Myeloid housekeeping
12 · Resident macrophage
17 · Membrane dynamics
15 · Endocytosis / TLR2
7 · Glucocorticoid resp
11 · Vesicle transport
13 · Actin cytoskeleton
8 · Myeloid differentiation
10 · Chromatin / epigenetic
6 · Vesicle / signaling
9 · Inflammatory myeloid
4 · Adhesion
Mac
6 · Noncoding RNA
10 · Complement / plasma
25 · FXR / bile-acid
2 · Oxidative phosphorylation
7 · Uncharacterized
13 · Bile-acid transport
1 · Plasma lipoprotein
27 · NF-kB immune activation
3 · Macrophage / immune
5 · Mitochondrial / metabolic
15 · Hypoxia
4 · Cholangiocyte secretory
21 · Inflammatory / oxidative
28 · Rho-GTPase / adhesion
23 · Mesenchymal / EMT
8 · Nutrient sensing
14 · Cell adhesion / actin
9 · Junction / autophagy
20 · Glucocorticoid / stress
24 · Cytoskeleton / FGFR2
18 · CFTR / ion transport
19 · Cytoskeleton / TGFB
16 · Mixed metabolic
17 · cAMP / PDE signaling
22 · Progenitor / Notch
11 · Ductal morphogenesis
26 · Adhesion / polarity
12 · Ciliary / apical
Chol
1 · Plasma protein
7 · Chromatin remodeling
5 · T-cell activation
6 · Rho-GTPase
2 · T-cell signaling
10 · Translation / MYC
4 · Translation / ribosome
11 · Immediate-early
9 · NK / cytotoxic
8 · Memory / MAIT
3 · Cytotoxic / MHC-I
T
Module score (centered)
1
0.5
0
-0.5
-1
GLS stage β
nominal 95% CI
117-program
Healthy
Steatosis
Steatohepatitis
-2
-1
0
1
GLS stage β